

Statistics for Quants

From Data to Decisions, Not p-Values

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December 26, 2025

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All three are independent. Backtest shows Fit, not Hedge quality.

Inference Quality Checklist (30-second):

$$\text{Score} = \underbrace{\frac{\sigma_{\text{OOS}}}{\sigma_{\text{IS}}}}_{< 1.5?} \times \underbrace{\frac{\text{Residual PnL}}{\text{Total PnL}}}_{< 5\%?} \times \underbrace{N_{\text{params}} / \sqrt{T}}_{< 1?}$$

If Score > 2 → Inference is weak

Module 2: Data Is Not IID (And Never Was)

Module Overview

What this module is about: This module explains why the IID (Independent and Identically Distributed) assumption—foundation of 95% of statistics textbooks—fails in markets and leads to catastrophic risk underestimation.

Why this matters: If you assume IID when data is dependent, your Sharpe ratios are lies, your VaR is wrong, and your hedges fail.

What confusion this clears:

- Why CLT misleads in finance
- Why volatility clusters
- Why tail events arrive in bursts

After this module, you should be able to answer: “Why does historical volatility underestimate future risk?”

2.1 The IID Fiction vs Market Reality

Assumption	Academic Stats	Market Reality
Independent	No memory	Vol clustering: high vol → more high vol
Identical	Same distribution	Regime shifts: crisis vs calm distributions differ
Stationary	Parameters fixed	Drift/vol evolve: $r_d - r_f, \rho$ change
Normal	CLT applies	Fat tails: jumps, kurtosis > 3

Table 6: IID assumption failure in financial data

Core Formula

$$\text{IID: } X_t \perp\!\!\!\perp X_{t-1}, \quad \mathbb{E}[X_t] = \mu, \text{ Var}(X_t) = \sigma^2$$

Core Formula

$$\text{Reality: } \sigma_t^2 = \omega + \alpha X_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (\text{GARCH})$$

The "PTSD Market" Mnemonic:

- P = Persistence (vol clustering)
- T = Thick tails (kurtosis > 3)
- S = Serial correlation (memory)
- D = Drift (non-stationarity)

Interview line: "Markets have PTSD, not amnesia."

"Interview: 'I use GARCH for baseline, overlay jump filter for tails.'"

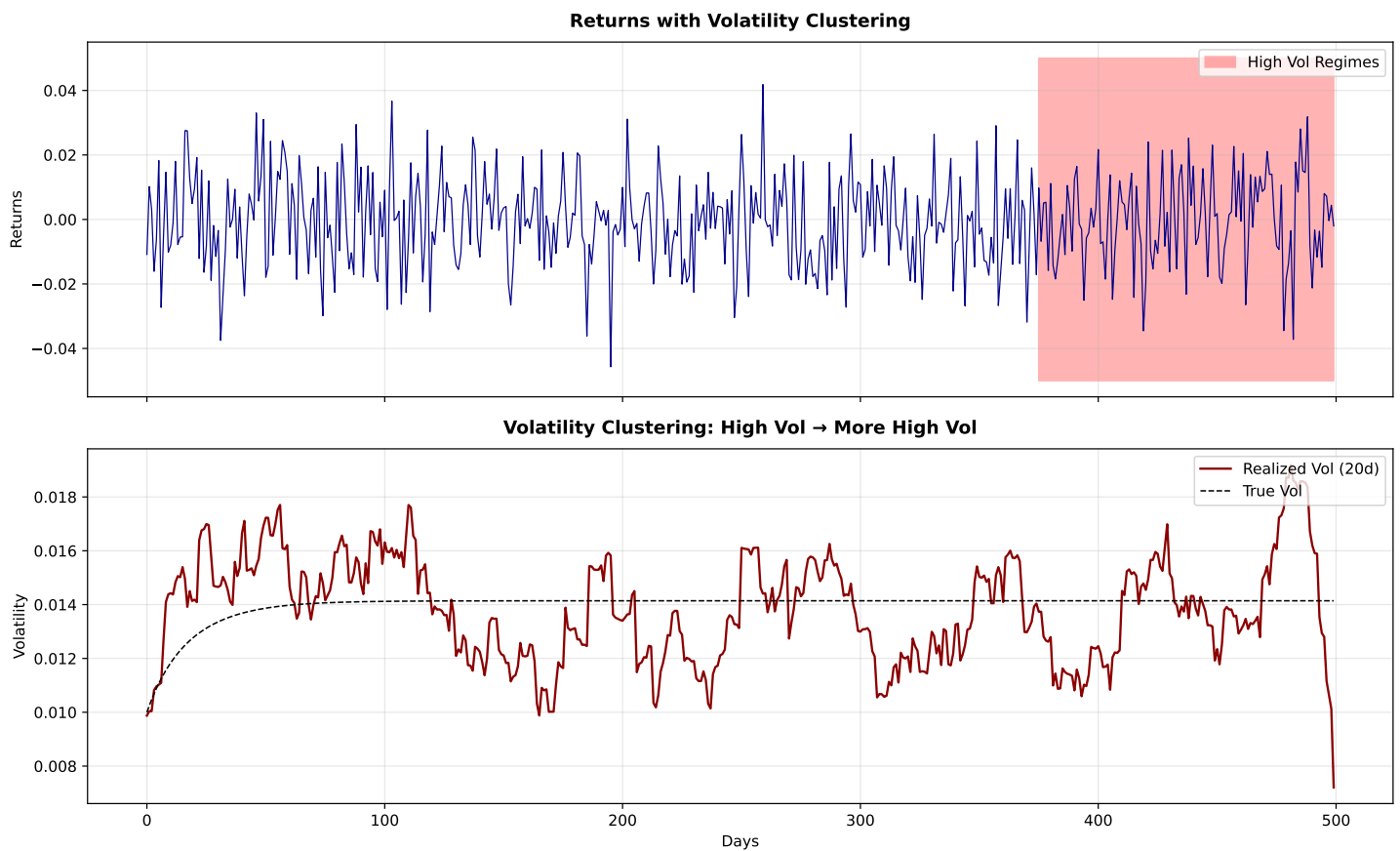


Figure 5: GARCH Volatility Clustering: Market Reality

6.3 Quick Interview Cheat Sheet — Module 6

Module 6: Vol = Stochastic Process

Core Concept: Volatility is not a parameter—it's a **conditional forecast**.

Must-Know:

- Realized vol is backward, conditional is forward
- GARCH persistence $\alpha + \beta \approx 0.97$
- Vol of vol ζ drives option smile convexity

Interview Triggers:

- "GARCH vs realized?" → "GARCH smooths, realized jumps"
- "Why GARCH fails?" → "Parameters not robust, slow crisis response"
- "How to trade vol?" → "Sell realized, buy conditional (var swap)"

Desk Wisdom:

- Use GARCH for term structure, not tails
- Overlay jump-diffusion for policy events

Common Trap: "Don't use GARCH σ_t as if it's observed." (It's model output)

Memory Aid: "Realized = rearview, conditional = GPS, implied = market's GPS."

Interviewer Is Testing: Can you forecast **distribution** not just point estimate?

- Histogram multimodal? → Mixture model

3. Level 3: Dependence Structure (15 min)

- ACF(returns) → Mean reversion?
- ACF(returns²) → Vol clustering?
- PACF → AR order?
- Ljung-Box → Serial correlation?
- ADF → Unit root?

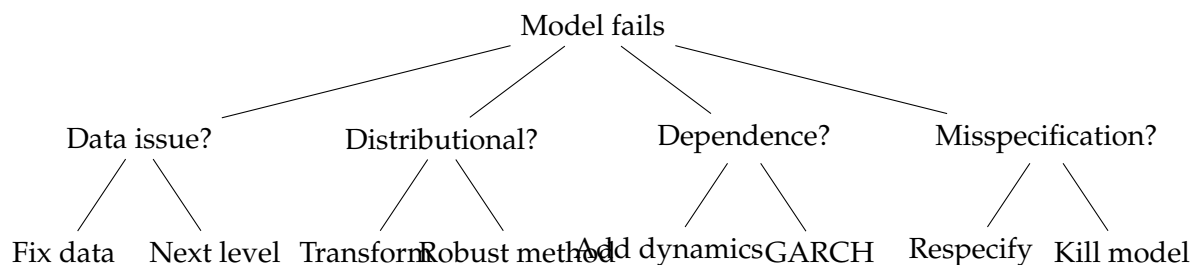
4. Level 4: Model Diagnostics (20 min)

- Residual plot → Pattern? (Misspecification)
- Residual ACF → Autocorr? (Add AR/MA)
- Residual² ARCH-LM → Vol clustering? (Use GARCH)
- QQ plot (residuals) → Non-normal? (Robust SE)
- Rolling → Stable? (Parameter constancy)

5. Level 5: Out-of-Sample Validation (30 min)

- Walk-forward backtest → Sharpe OOS vs IS
- Parameter stability → $_t$ vs $_{t-60}$
- Residual PnL → $|| < 5\%$
- VaR backtesting → Kupiec + Christoffersen
- Stress testing → Max drawdown $< 3 \times$ expected?

The Decision Tree



Mental Models: Think Like a Bayesian Statistician

Mental Model 1: "Everything Is a Distribution"

DON'T think: $\hat{\beta} = 0.5$

DO think: $\hat{\beta} \sim \mathcal{N}(0.5, 0.1^2)$

- Parameter → Distribution
- Forecast → Predictive distribution
- Risk → Tail of distribution

Mental Model 2: "Uncertainty Compounds"

$$\text{Total Uncertainty}^2 = \underbrace{\text{Parameter}^2}_{\text{Estimation}} + \underbrace{\text{Model}^2}_{\text{Specification}} + \underbrace{\text{Market}^2}_{\text{Volatility}}$$